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003601

May 2023

B.Tech. (CE/IT) - VI SEMESTER

Intelligent Systems (PCC-CS-601)

ime : 3 Hours] [Max. Marks : 75

istructions :

1. *It is compulsory to answer all the questions (1.5 marks each) of Part-A in short.*
2. *Answer any four questions from Part-B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART–A

1. (a) What are the applications of Artificial Neural Network? (1.5)

(b) Define Stochastic Annealing. (1.5)

(c) List the advantages, disadvantages, time and space complexity of Breadth first search algorithm. (1.5)

(d) Define Strong Artificial Intelligence. (1.5)

(e) What do you understand by Semantic Net? Draw the semantic net for the following given statement. (1.5)

- "Cycle is a two wheeler and it is a moving vehicle. Vehicle needs an engine, a fuel system to sustain the engine running, an electric system for its lights, horn and breaks and also used by children for racing and enjoying".

(f) Explain Turing Test In AI. (1.5)

(g) What are the Task domains of AI? (1.5)

(h) What do you mean by Problem reduction in AI? (1.5)

(i) Represent the following statements using Thematic Role Frame. (1.5)

- The cookies were eaten in the kitchen under the table by the baker.

(j) Differentiate between Supervised and Unsupervised learning algorithm. (1.5)

PART–B

2. (a) What is neural network? How Artificial Neural Networks is derived from Natural Biological neuron? (10)

(b) What is RBFNN? How it is different from Multilayer Feed-Forward NN? (5)

3. (a) Draw and Explain the architecture of Fuzzy Neural System. (5)

(b) What is predicate logic? Why it is preferred over propositional logic? Explain with the help of example how predicate logic is used as one of the technique for representing knowledge in computer system? (10)

. Explain Iterative deepening method along with the advantages over the other Brute Force Search methods. Also Compare their time and space Complexity. (15)

(a) What is Fuzzy logic? Why it is important? Expl
Fuzzy Logical operations, and support and core
Fuzzy set?

(b) What is Genetic algorithm? When it is applicab
Briefly, explain how it can be applied on Travelli
Salesman problem. (

6. (a) Explain Best First Search technique with example a
algorithm. Also explain why it is preferred over h
climbing technique? (1

(b) Give a brief idea about blackboard architecture. (

7. Explain the following : (1

(a) Bayesian Reasoning and Bayesian Network.

(b) Dempster Shafer Theory of Evidence with derivatio

(c) Induction Learning.

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003601

August/September 2022
B.Tech. (CE/IT) VI SEMESTER
Intelligent Systems (PCC-CS-601)

ne : 3 Hours] Max. Marks : 75

structions :

1. *It is compulsory to answer all the questions (1.5 marks each) of Part-A In short.*
2. *Answer any four questions from Part-B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART-A

1. (a) What is an Intelligent System? (1.5)
   (b) Explain supervised and unsupervised learning. (1.5)
   (c) What is a Recurrent network? (1.5)
   (d) What do you mean by membership value in fuzzy logic? (1.5)
   (e) What is an iterative deepening search? (1.5)
   (f) What is Reasoning? What are its types? (1.5)
   (g) What are the advantages of predicate logic over propositional logic? (1.5)

(b) What are Evolutionary Algorithms? Give exa

PART-B

2. (a) Explain the Backpropagation network in detail.
   (b) Differentiate between ANN and biological ne network.

3. (a) Explain the Genetic Algorithm in detail along various operators.
   (b) Discuss the various ways of knowledge representa

4. (a) What is a heuristic search? How it is better? Expl A* algorithm with an example.
   (b) What is Hill climbing? What are its limitations?

5. (a) What is an Expert System? Explain its architecture detail.
   (b) Show that $(\sim W \vee R)$ is a logical consequence of a s $\{\sim P \vee Q, \sim Q \vee R, \sim W \vee P\}$ using the resolutio refutation method.

(a) Explain Bayesian reasoning in detail. (10)

(b) Consider a set $\{\sim(P \vee Q), (Q \vee R), \sim(\sim P \wedge R)\}$ of formulae. Show that the set is inconsistent using the semantic tableau method. (5)

Write short note on :

(a) Blackboard Architecture.

(b) Inductive and Deductive reasoning.

(c) Semantic nets and Frames. (3×5=15)

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May 2024

B.Tech. (CE/CE/Hindi)/IT) (Sixth Semester)

Intelligent Systems (PCC-CS-601)

Time : 3 Hours] [*Maximum Marks* : 75

Note : It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other.

Part A

1. (a) What are the various uses of the intelligent system ? 1.5

(b) What do you mean by Perceptron ? 1.5

(c) List out the application of the Genetic algorithm. 1.5

(d) Differentiate Between crisp logic and fuzzy logic. 1.5

(e) What is a Heuristic search ? Explain with an example. 1.5

(f) What are the applications of iterative deepening search ? 1.5

P.T.O.

Differentiate between database and knowledge base. 1.5

What is the semantic net ? 1.5

State Dempster-Shafer's Theory. 1.5

What is Bayesian reasoning ? 1.5

Part B

a) Appraise are the advantages and disadvantages of backpropagation network. 10

(b) Compare recurrent neural networks and feed-forward neural networks. 5

3. (a) Explain every step of the Genetic algorithm in detail. 10

(b) What is the main difference between probability and fuzzy logic ? 5

4. (a) Explain the AO* Algorithm with the help of an example. 10

(b) Compare breadth-first search and depth-first search. 5

5. (a) Explain different techniques of knowledge representation with examples. 10

(b) Draw the diagram of the blackboard system and explain each component. 5

(a) Write a note on statistical learning and induction learning. 10

(b) What is Certainty factor in AI ? 5

. Draw and describe the architecture of the expert system. 15

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May 2025

B.Tech. (Sixth Semester)

Intelligent Systems (PCC-CS-601)

Time : 3 Hours] [*Maximum Marks : 75*

Note : It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other.

Part A

1. (a) What are the application areas of an intelligent system ?

   (b) What are the different types of learning in ANN ?

   (c) Explain the terms core and height of a fuzzy set.

   (d) Explain how IDDFS is better than BFS or DFS.

   (e) Explain the water-jug problem.

P.T.O.

(f) Discuss the various ways of knowledge representation.

(g) Explain induction learning.

(h) What are Evolutionary Algorithms ? Give example.

(i) Discuss, why a backpropagation network is required ?

(j) What do you mean by problem reduction in AI ? 1.5×10=15

Part B

2. (a) Differentiate between BNN and ANN. 5

(b) Why Recurrent Neural Network is preferred in AI, how many layers can be there in RNN ? Write the disadvantages of RNN if any. 10

3. (a) Why fuzzy logic was introduced ? Design a fuzzy set for the human age. 5

(b) What is a Genetic Algorithm ? and why is it used ? Explain the various Genetic Operations. 10

4. (a) Show that $P \rightarrow S$ is provable from the set $\{P \rightarrow Q, Q \rightarrow R, R \rightarrow S\}$ using semantic tableaux method. 5

(b) Explain the A* algorithm. How is it different from Hill Climbing ? Why is it called an optimized algorithm ? 10

5. (a) Show that $(\sim W \vee R)$ is a logical consequence of a set $\{\sim P \vee Q, \sim Q \vee R, \sim W \vee P\}$ using the resolution refutation method. 5

(b) What is an Expert System ? Explain its architecture in detail along with inferencing techniques. 10

6. (a) What is Reasoning ? What are its types ? Also, explain inductive and deductive reasoning with examples. 5

(b) Explain Dempster Shafer's Theory of Evidence along with its advantages and disadvantages. 10

7. Write short notes on the following : 15

(a) Blackboard architecture

(b) Semantic nets and Frames

(c) AO* algorithm

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